C73-15: A closed-certificate characterization for every fixed mirror-descent rule

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The characterization is infinitary. No finite recognition algorithm, step-size-independent potential criterion, or familiar-game-class list is claimed. The construction retains the arbitrary specified regularizers and positive schedule, including allowed boundary iterates. The fixed mirror-descent family, schedule, and universal-interior-initialization target are the specialization C73-15 of the broader source agenda; the result does not classify every learning dynamic in that agenda.

1 Exact target and the form of the answer

Fix $d \geq 1$ players and nonempty finite action sets S_i, $i = 1, \ldots, d$, $X_i = \Delta(S_i)$, and $X = \prod_i X_i$. The fixed rule has proper lower-semicontinuous strictly convex regularizers $h_i : X_i \to \mathbb{R} \cup \{+\infty\}$, finite and differentiable on relint X_i, and steps $\eta_t > 0$. For a game's payoff array U, write u_i^U for its multilinear extension and $v_i^U(x)_a = u_i^U(a, x_{-i})$. The exact simultaneous update is

$$x_i^{t+1} = \arg\max_{p \in X_i} \left\{ \eta_t \langle v_i^U(x^t), p \rangle - h_i(p) + h_i(x_i^t) + \langle \nabla h_i(x_i^t), p - x_i^t \rangle \right\}. \tag{1}$$

We retain the canonical assumption that these updates are uniquely defined and remain in the declared gradient domain from every interior initialization. No assertion is made about a payoff array for which this assumed rule is undefined.

Our answer associates to U countable trees $T_{a,b}(U)$, $a, b \in \mathbb{N}_{>0}$. Node validity is a finite closed feasibility problem: it uses fixed regularizer epigraphs, regularizer evaluations at interior probes and interpolated interior points, the specified steps, and explicit payoff polynomials. The interpolated points depend on the finite feasibility variables; the regularizers themselves remain fixed rule data. It contains no Nash set, convergence assertion, infinite trajectory, or gradient evaluation. The condition on the payoff array is

$$\boxed{T_{a,b}(U) \text{ is well-founded for every } a, b \geq 1.} \tag{2}$$

Equivalently each tree admits a strictly decreasing countable-ordinal ranking. The full definition is in Section 4. Sections 2–3 explain why its finite constraints encode interior-ray slopes and exact updates without imposing Fréchet differentiability or continuity of the boundary gradient.

2 Closed certificates for interior-ray slopes

For each i, fix an enumeration $(q_{i,\ell})_{\ell \geq 1}$ of the rational points of relint X_i. It is dense in relint X_i, hence in X_i. All numbers $h_i(q_{i,\ell})$ are finite and fixed before testing U. For a singleton simplex repeat its unique point.

We normalize a declared gradient by choosing its representative g with $\sum_{s \in S_i} g_s = 0$. Adding a constant to all gradient coordinates has no effect on a Bregman divergence or on (1). The proof uses only the defining directional property of a gradient on each interior ray:

$$\lim_{\theta \downarrow 0} \frac{h_i(x + \theta(q - x)) - h_i(x)}{\theta} = \langle \nabla h_i(x), q - x \rangle \qquad (q \in \text{relint } X_i). \tag{3}$$

This holds both for a Fréchet gradient and for a linear finite-domain Gâteaux gradient. We do not assert a uniform expansion at a boundary point, or finite regularizer values in a whole boundary neighborhood. A selected subgradient unrelated to these directional derivatives would be a different notion from the declared gradient.

Lemma 1 (Dense interior-ray certificate)**.** *Let P be a finite-dimensional simplex and let $h : P \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and convex, finite on* relint P*. Fix dense interior probes (q_ℓ). For $x \in P$, $c \in \mathbb{R}$, and a normalized vector g, the following are equivalent:*

1. *$h(x) = c$, and the derivative along each fixed probe ray exists and satisfies*

$$\lim_{\theta \downarrow 0} \frac{h(x + \theta(q_\ell - x)) - h(x)}{\theta} = \langle g, q_\ell - x \rangle \qquad (\ell \geq 1). \tag{4}$$

2. *$c \geq h(x)$, and, for every ℓ,*

$$E_\ell := h(q_\ell) - c - \langle g, q_\ell - x \rangle \geq 0; \tag{5}$$

for every pair of integers $\ell, k \geq 1$ there is an integer $N_{\ell,k} \geq 1$ such that, writing $z_{\ell,k} = (1 - 1/N_{\ell,k})x + q_\ell/N_{\ell,k}$,

$$0 \leq h(z_{\ell,k}) - c - \frac{\langle g, q_\ell - x \rangle}{N_{\ell,k}} \leq \frac{\|q_\ell - x\|}{kN_{\ell,k}}. \tag{6}$$

The conclusion is the displayed ray identity. It does not assert full Fréchet differentiability at an arbitrary coded boundary point.

Proof. A convex function finite on a relative open set is continuous there. One elementary justification is to enclose a small neighborhood in a simplex whose vertices lie in that set. Convexity bounds the function above by the largest vertex value; reflection about the center then bounds it below. Applying these bounds on a slightly larger neighborhood gives bounded secant slopes, hence continuity on the smaller one.

Assume the second condition. The epigraph inequality makes $h(x)$ finite. Interior continuity extends (5) to all interior q. Choose one fixed interior q^0 and put $z_\theta = (1 - \theta)x + \theta q^0$, $0 < \theta \leq 1$. Convexity and lower semicontinuity give

$$\lim_{\theta \downarrow 0} h(z_\theta) = h(x) : \quad h(z_\theta) \leq (1 - \theta)h(x) + \theta h(q^0), \quad h(x) \leq \liminf_{\theta \downarrow 0} h(z_\theta).$$

Letting $\theta \downarrow 0$ in (5) at z_θ yields $c \leq h(x)$. Thus $c = h(x)$.

The supporting inequality extends to any finite-value boundary point p by applying it to $(1 - \theta)p + \theta q^0$ and using the same limit identity. It is automatic when $h(p) = +\infty$. Consequently

$$h(p) - h(x) - \langle g, p - x \rangle \geq 0 \qquad (p \in P). \tag{7}$$

For a fixed probe q_ℓ, convexity makes its secant quotient

$$\phi_\ell(\theta) = \frac{h(x + \theta(q_\ell - x)) - h(x)}{\theta}, \qquad 0 < \theta \leq 1,$$

nondecreasing as a function of θ. Its limit at zero is therefore its infimum, possibly an extended real number before the certificate is used. By (7) every quotient is at least $\langle g, q_\ell - x \rangle$. Multiplying (6) by $N_{\ell,k}$ gives

$$\phi_\ell(1/N_{\ell,k}) \leq \langle g, q_\ell - x \rangle + \|q_\ell - x\|/k.$$

Hence the infimum is at most the right side for every k, and equals $\langle g, q_\ell - x \rangle$. This proves (4), without uniformity over directions.

Conversely, (4) and secant monotonicity imply (5). For each fixed ℓ, k, the quotients along $\theta = 1/N$ decrease to the finite value in (4), so some integer N makes their excess at most $\|q_\ell - x\|/k$. When $q_\ell = x$ the excess is identically zero. This is (6). □

At a point where the original rule declares a gradient, (3) and (4) identify the certificate vector with its normalized representative. Indeed their inner products with every $q_\ell - x$ agree. By density these vectors span the affine tangent space, and normalization removes the only ambiguity, an all-coordinate constant. We will use this identification only at points already identified as actual iterates of the declared rule.

Lemma 2 (Dense-probe optimizer certificate)**.** *Suppose $p \in P$, $c = h(p) \in \mathbb{R}$, and y is a finite vector. Then p maximizes $\langle y, z\rangle - h(z)$ over P if and only if*

$$h(q_\ell) - c - \langle y, q_\ell - p\rangle \geq 0 \qquad (\ell \geq 1). \tag{8}$$

Proof. The inequalities extend first to the interior by continuity, then to finite-value boundary points by the convex-segment limit used above. Infinite-value points cannot improve the objective. The extended inequalities are exactly the maximization assertion. The reverse implication is immediate. □

These lemmas do not assume continuity of the boundary gradient. For a finite list of probes, (5) is closed. With an integer $N_{\ell,k}$ fixed, the interpolated point lies in the relative interior for every $x \in P$. Every coordinate of that point is at least the corresponding positive coordinate of $q_\ell/N_{\ell,k}$, so all such points belong to a compact subset of the relative interior. Thus $h(z_{\ell,k})$ is continuous in x, and both inequalities in (6) are closed. The epigraph condition $c \geq h(x)$ is closed by lower semicontinuity. These closure facts are the reason for retaining ray-specific denominators as integer node labels instead of assuming that finite mirror-descent trajectories depend continuously on their initialization.

3 Payoff polynomials and compact initial cores

Define pure-deviation regrets

$$R^U_{i,s}(x) = u^U_i(s, x_{-i}) - u^U_i(x), \qquad R^U(x) = \max_{i,s} R^U_{i,s}(x). \tag{9}$$

They are explicit polynomials in mixed-action probabilities, with coefficients from U. For example

$$u^U_i(x) = \sum_{z \in \prod_j S_j} U_i(z) \prod_j x_{j,z_j}, \qquad v^U_i(x)_s = \sum_{z_{-i}} U_i(s, z_{-i}) \prod_{j \neq i} x_{j,z_j}. \tag{10}$$

For an integer $a \geq 1$, let

$$K_a = \{x \in X : x_{i,s} \geq 1/a \text{ for every } i, s\}. \tag{11}$$

Each K_a is compact, possibly empty, and $\bigcup_{a \geq 1} K_a = \operatorname{relint} X$. Thus compact cores cover precisely all permitted initial profiles, even arbitrarily close to the boundary.

4 The finite node test and the payoff criterion

Fix $a, b \geq 1$. A node of depth $r \geq 1$ has only finite discrete labels:

- strictly increasing integer times $1 \leq T_1 < \cdots < T_r$;
- player/action labels (i_j, s_j), one for each marked time;
- positive integer bounds B_t, $1 \leq t \leq T_r$;
- positive integer denominators $N_{i,t,\ell,k}$, $1 \leq i \leq d$, $1 \leq t \leq T_r$, $1 \leq \ell, k \leq r$.

The empty node is the root. Truncation from depth r to depth $r-1$ retains the earlier marked times and their labels, the bounds with $t \leq T_{r-1}$, and denominator labels with $t \leq T_{r-1}$, $\ell, k \leq r-1$. Extensions must retain all those labels unchanged. Therefore the set of possible nodes is countable. The continuous variables below are feasibility witnesses, not node labels.

A depth-r node is valid if the following finite system has a solution in variables $x^t \in X$, $c_{i,t} \in \mathbb{R}$, $g_{i,t} \in \mathbb{R}^{S_i}$, $1 \leq t \leq T_r$:

1. **Compact bounds and initial core:**

$$x^1 \in K_a, \qquad |c_{i,t}| \le B_t, \qquad |g_{i,t,s}| \le B_t, \qquad \sum_s g_{i,t,s} = 0. \tag{12}$$

2. **Epigraph and finite supporting probes:** for every i, t,

$$c_{i,t} \ge h_i(x_i^t), \qquad E_{i,t,\ell} := h_i(q_{i,\ell}) - c_{i,t} - \langle g_{i,t}, q_{i,\ell} - x_i^t \rangle \ge 0 \quad (1 \le \ell \le r). \tag{13}$$

3. **Finite interior-ray secants:** for every i, t and $1 \le \ell, k \le r$, put

$$z_{i,t,\ell,k} = \left(1 - \frac{1}{N_{i,t,\ell,k}}\right) x_i^t + \frac{q_{i,\ell}}{N_{i,t,\ell,k}}.$$

Require

$$0 \le h_i(z_{i,t,\ell,k}) - c_{i,t} - \frac{\langle g_{i,t}, q_{i,\ell} - x_i^t \rangle}{N_{i,t,\ell,k}} \le \frac{\|q_{i,\ell} - x_i^t\|}{k N_{i,t,\ell,k}}. \tag{14}$$

4. **Finite update probes:** for $1 \le t < T_r$, every i, and $1 \le \ell \le r$,

$$h_i(q_{i,\ell}) - c_{i,t+1} - \left\langle g_{i,t} + \eta_t v_i^U(x^t), q_{i,\ell} - x_i^{t+1} \right\rangle \ge 0. \tag{15}$$

5. **Marked payoff gaps:**

$$R_{i_j,s_j}^U(x^{T_j}) \ge 1/b \qquad (1 \le j \le r). \tag{16}$$

The tree $T_{a,b}(U)$ consists of its root and its valid nodes under the specified truncation. Validity is inherited by truncation, since all retained constraints occur among the child's constraints.

Every test is finite and closed in its real feasibility variables. In particular, for each fixed denominator label the argument of the regularizer in (14) ranges in a compact subset of its relative interior, where it is continuous. The payoff dependence occurs only in the polynomials (10), (15), and (16). Everything else belongs to the fixed rule and the fixed probe system. The root can be included even when K_a is empty; then it has no valid children and is well-founded.

Theorem 3 (Full payoff characterization)**.** *For every payoff array U for which the canonical fixed rule is well-defined and gradient-domain preserving as assumed, the following are equivalent:*

1. *For every $x^1 \in \operatorname{relint} X$, the exact simultaneous rule* (1) *satisfies* $\operatorname{dist}(x^t, \mathrm{NE}(U)) \to 0$.
2. *The payoff-array criterion* (2) *holds for the finite tests* (12)–(16).
3. *For every a, b, there is a countable ordinal $\alpha_{a,b}$ and a map $\rho_{a,b} : T_{a,b}(U) \to \alpha_{a,b}$ strictly decreasing from a node to each proper extension.*

5 Proof of the full equivalence

Regret and distance have the same vanishing criterion. The function R^U is continuous and nonnegative: a player's mixed payoff is a convex combination of its pure-action payoffs, so at least one of its pure deviations has nonnegative regret. Moreover $R^U(x) = 0$ if and only if every pure deviation is unprofitable, which is equivalent by linearity to $x \in \mathrm{NE}(U)$. The Nash set is nonempty by the finite-game existence theorem [2] and is closed in compact X.

For any sequence $z_t \in X$, if its distance to the Nash set tends to zero, uniform continuity of R^U implies $R^U(z_t) \to 0$. Conversely, if $R^U(z_t) \to 0$ but its Nash distance does not, a subsequence with distance at least some $\varepsilon > 0$ has a compact subsequential limit z. Continuity gives $R^U(z) = 0$, hence $z \in \mathrm{NE}(U)$, a contradiction.

Failure of convergence produces an infinite branch. Suppose an actual trajectory from interior x^1 fails to approach the Nash set. Choose a with $x^1 \in K_a$. Its nonnegative regrets fail to tend to zero,

so for some $b \geq 1$ there are strictly increasing times T_j with $R^U(x^{T_j}) \geq 1/b$. Choose a player and a pure action attaining this maximum at each such time.

Put $c_{i,t} = h_i(x_i^t)$, and let $g_{i,t}$ be the normalized relative gradient supplied by the declared rule. All these numbers are finite. Choose integer B_t exceeding their absolute values. Property (3) and Lemma 1 supply integer denominators $N_{i,t,\ell,k}$ for every date, probe, and precision. Fix each such choice once. Lemma 2 applied to (1) gives all the update inequalities. At depth r, the actual finite prefix through T_r satisfies (12)–(16). Its labels extend the preceding node. These nodes form an infinite branch of $T_{a,b}(U)$, so (2) fails.

An infinite branch produces one exact bad trajectory. Conversely, suppose $T_{a,b}(U)$ has an infinite branch. Its marked times strictly increase and therefore tend to infinity. Along the branch every bound B_t is eventually assigned and never changed. Each denominator $N_{i,t,\ell,k}$ is also eventually assigned and never changed.

At each depth choose one finite feasibility witness. Extend it beyond its last date by arbitrary points of the coordinate boxes

$$\prod_{t\geq 1}\left[X \times \prod_i [-B_t, B_t] \times \prod_i [-B_t, B_t]^{S_i}\right]. \tag{17}$$

These boxes are all compact and nonempty. Extract a subsequence converging in the first date's coordinates, then a subsequence converging in the second date's coordinates, and so on. The diagonal subsequence converges coordinatewise to arrays $(x^t, c_{i,t}, g_{i,t})_{t\geq 1}$. Each constraint from any fixed node involves only finitely many coordinates and is closed; all sufficiently deep witnesses satisfy it. Its limit therefore satisfies that constraint. This proves simultaneously every constraint on the branch, without assuming continuity of a mirror-descent trajectory map.

For fixed i, t, each probe ℓ and precision k eventually occurs in a branch node. Hence the limit arrays satisfy the full conditions of Lemma 1, with the same fixed denominator for each pair. It follows that $c_{i,t} = h_i(x_i^t)$, and the vector $g_{i,t}$ has the certified linear slopes on every fixed interior probe ray. We do not infer a full Fréchet gradient at an arbitrary coded boundary point. All update probes eventually occur as well, so Lemma 2 identifies x_i^{t+1} as the maximizer with score $g_{i,t} + \eta_t v_i^U(x^t)$.

We now identify the sequence with the actual declared rule by induction. The first point belongs to $K_a \subseteq \operatorname{relint} X$, where the original gradient exists. Its directional slopes satisfy (3). The certified slopes (4) match them on every probe ray, and the dense directions span the tangent space. Therefore $g_{i,1}$ is the original normalized gradient. The optimizer just identified is the original (1) optimizer and is unique, so x^2 is the actual next iterate. The canonical admissibility assumption puts x^2 in the original declared gradient domain. At this known actual point, (3), the certified ray identities, and dense spanning again identify $g_{i,2}$ with the normalized original gradient. The same argument proves the assertion at every date.

This induction uses the original gradient-domain preservation; it does not deduce that arbitrary points admitted by a ray certificate belong to that domain. It permits the original finite-domain Gâteaux interpretation as well as the Fréchet interpretation, and does not require the regularizer to be finite near an allowed boundary iterate.

Finally all marked inequalities (16) survive in the limit, so $R^U(x^{T_j}) \geq 1/b$ at infinitely many dates. This trajectory does not approach the Nash set. We have proved the equivalence of assertions 1 and 2.

The ordinal version. For a countable well-founded tree, well-founded recursion defines

$$\rho(v) = \sup\{\rho(w) + 1 : w \text{ is an immediate child of } v\}, \qquad \sup \varnothing = 0.$$

Each rank is a countable ordinal, since each node has at most countably many children and a countable supremum of countable ordinals is countable. The tree has countably many nodes, so all its ranks are bounded by a single countable ordinal. The ranks strictly decrease along extensions. Conversely an infinite branch would give an infinite strictly decreasing sequence of ordinals, which is impossible. This proves assertions 2 and 3 equivalent and completes Theorem 3.

6 Scope

The original regularizers and every original step are retained. No entropy, summability, small-step, strong-convexity modulus, potential structure, gradient Lipschitz bound, or boundary-gradient continuity is added. The interior initial set is covered by all compact cores, rather than replaced by one selected core. The update is unique by hypothesis, so there is no additional tie-breaking convention. All players use the same old profile in the simultaneous payoff vectors of (15). The target is distance to the Nash set, not action convergence, time-averaged convergence, or convergence to one equilibrium.

The condition is a payoff-structure criterion in a closed-certificate sense: it consists of explicitly specified finite constraints with payoff-polynomial coefficients, followed by well-foundedness or an ordinal ranking. Real trajectory coordinates appear only as finite feasibility variables and are eliminated from the countable tree itself. Neither exact gradient evaluation nor infinite orbit quantification defines node validity. The supporting regularizer probes and ray-specific secant certificates are needed because ordinary compactness applied to unclosed update graphs would be insufficient.

This is an infinitary representation. An arbitrary fixed regularizer or step sequence need not have an effective finite encoding. Even decidability of each finite node test would not decide well-foundedness of the entire tree. Accordingly no recognition algorithm or running-time bound is claimed. For general regularizers these finite tests need not be polynomial or semialgebraic: fixed epigraph membership is part of the rule's data.

The tree has countably many possible waiting times and secant-denominator labels. In general, countable well-founded trees need not have finite height; the criterion itself imposes no uniform finite horizon. No uniform bound on the number of bad dates or natural-number ranking is inferred. Countable ordinals retain the full pointwise-in-initialization convergence property.

7 A boundary test for pointwise ray certificates

On the three-action simplex write the first two probabilities as x, y, so $x, y \geq 0$ and $x+y \leq 1$. Consider

$$h(x,y) = x^2 + y^2 + \frac{y^2}{\sqrt{x}} \quad (x>0), \qquad h(0,0) = 0, \qquad h(0,y) = +\infty \quad (y>0). \tag{18}$$

It is proper and lower semicontinuous. The Hessian of $y^2/\sqrt{x}$ for $x>0$ has nonnegative diagonal entries and determinant $y^2/(2x^3) \geq 0$, so this term is convex there. Nonnegativity gives lower semicontinuity at the origin, and the term $y^2/\sqrt{x}$ tends to infinity when $x \downarrow 0$ and y approaches a positive boundary value. For a segment from the origin to a finite-value point with $x>0$, $h(tx,ty) = t^2(x^2+y^2) + t^{3/2}y^2/\sqrt{x} \leq th(x,y)$ for $0 \leq t \leq 1$; the interior Hessian handles all other finite-value segments, so the stated extension is convex. Adding x^2+y^2 makes h strictly convex, and it is smooth on the relative interior.

At the vertex $(0,0)$, every finite-domain ray has linear directional gradient zero: for a fixed interior probe $q = (a, b, 1-a-b)$, put $d = q - (0,0,1) = (a, b, -a-b)$, using the ambient Euclidean norm. Then

$$\frac{h(\theta a, \theta b)}{\theta} = \theta(a^2+b^2) + \sqrt{\theta}\,\frac{b^2}{\sqrt{a}} \longrightarrow 0.$$

Yet $h(r^4, r) = r^8 + r^2 + 1$ does not even tend to $h(0,0)$. A uniform Fréchet certificate would exclude this declared weak boundary gradient. The ray certificate includes it: take $c=0$, $g=0$. The supporting probes are nonnegative, and, for each fixed q and k, choose a sufficiently large integer N so that

$$Nh(a/N, b/N) = \frac{a^2+b^2}{N} + \frac{b^2}{\sqrt{aN}} \leq \frac{\|d\|}{k}.$$

This is exactly (6), with no uniformity over probes required.

For one player, $\eta_1 = 1$, payoff vector $(7/8, -7/2, 0)$, and the interior profile $(1/16, 1/4, 11/16)$, the regularizer gradient representative is $(-15/8, 5/2, 0)$. The next optimization score is $(-1,-1,0)$, so its unique optimizer is $(0,0,1)$: every other point has strictly negative score minus the nonnegative regularizer. If the declared boundary gradient is the finite-domain linear directional gradient zero, the

following unit-step optimizer is $(7/16, 0, 9/16)$. The ray certificate permits this exact prefix. We do not use this test as a replacement for the full theorem or claim a separate all-initialization convergence result from this prefix.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] Martin Bichler, Davide Legacci, Panayotis Mertikopoulos, Matthias Oberlechner, and Bary Pradelski. *Characterizing the Convergence of Game Dynamics via Potentialness*, arXiv:2503.16285v1, submitted 20 March 2025. `https://arxiv.org/html/2503.16285v1`. Sections 3–4 provide game/potentialness context; Section 6 states the general classification agenda.

[2] J. F. Nash, Jr., *Equilibrium points in n-person games*, Proceedings of the National Academy of Sciences USA **36**(1), 48–49 (1950), DOI: 10.1073/pnas.36.1.48. `https://pmc.ncbi.nlm.nih.gov/articles/PMC1063129/`.